SENSE- A CONTEXT SENSITIVE DEVICE FOR MONITORING CHILDREN WITH SPECIAL NEEDS

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(S / 18 / 338)

for the partial fulfillment for the requirement for the course of

CS 412 Mobile and Pervasive Computing

To the **Department of Statistics and Computer Science** of the

FACULTY OF SCIENCE
UNIVERSITY OF PERADENIYA
SRI LANKA
2023

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INTRODUCTION

Self-monitoring is a fundamental aspect of human behavior. Unlike adults, most young children cannot self-monitor. A young child's response to dissatisfaction is shown through screaming or crying. While most children learn to self-monitor as they age, children with developmental disabilities may not master these skills without support and instructions. These special needed children may have physical, cognitive, and emotional conditions that require continuous, context aware supervision. For special needed children especially those with Autism, adding sensory toys and tools with them can tremendously help with self-monitoring, self-regulate and calm down when they overworked or overwhelmed with continuous monitoring by the parent and guardian.

With the advancements in mobile and pervasive computing, Edge AI and sensor technologies, development of a comprehensive solution to address the needs of a child with special needs with continuous monitoring can be done. A context-sensitive mobile device, worn by the child can sense and process environmental and emotional conditions, and alert and provide notifications on the guardian's device in real time. The device offers personalized monitoring system ensuring safety of the child, health management, emotional management,

The device would monitor environmental conditions like temperature, humidity and noise as well as track the child's pulse rate, oxygen levels. Also, it would keep parents informed about the child's location and movement patterns and routine by generating periodic reports that can aid in managing chronic medical conditions, physical disabilities. By integrating AI and mobile technologies, the device can act autonomously detecting anomalies, performing predictions, providing suggestions and alerting parents, guardians or caregivers when immediate action is necessary.

MOTIVATION

Children with special needs (Autism, Down Syndrome or other cognitive and physical disabilities) often face challenges in communicating their needs or discomfort. This puts parents, guardians a significant burden. While wearable technology and mobile devices are used in health monitoring applications, few solutions are tailored specifically for children with special needs. The aim of the project is to develop a context-sensitive device that collect and interprets real-time data ensuring that both parent and the child have the support they need, enhancing the child's quality of life while allowing parents to respond more effectively to immediate concerns.

OBJECTIVES

The primary objectives of this project are as follows

- Design a wearable, context-sensitive device for children with special needs that continuously monitor, track environmental and physiological conditions.
- Real time data collection on temperature, humidity, noise levels, pulse rate, oxygen saturation, and the child's location and movement.
- Integrating AI based analysis for generating insights on development delays, behavior, speech, self-care abilities, interacting with environment and academic progress.
- Providing timely notifications and alert system with reports to parents.
- Ensure medication through regular reminders and health status updates.

APPROACH

The project involves building a practical and reliable wearable device with mobile/ computer pairing that can provide comprehensive monitoring while being energy efficient. By using context-sensitive computing, we can create a system that adapts its behavior based on the child's health, emotional wellbeing, physical wellbeing and environmental conditions.

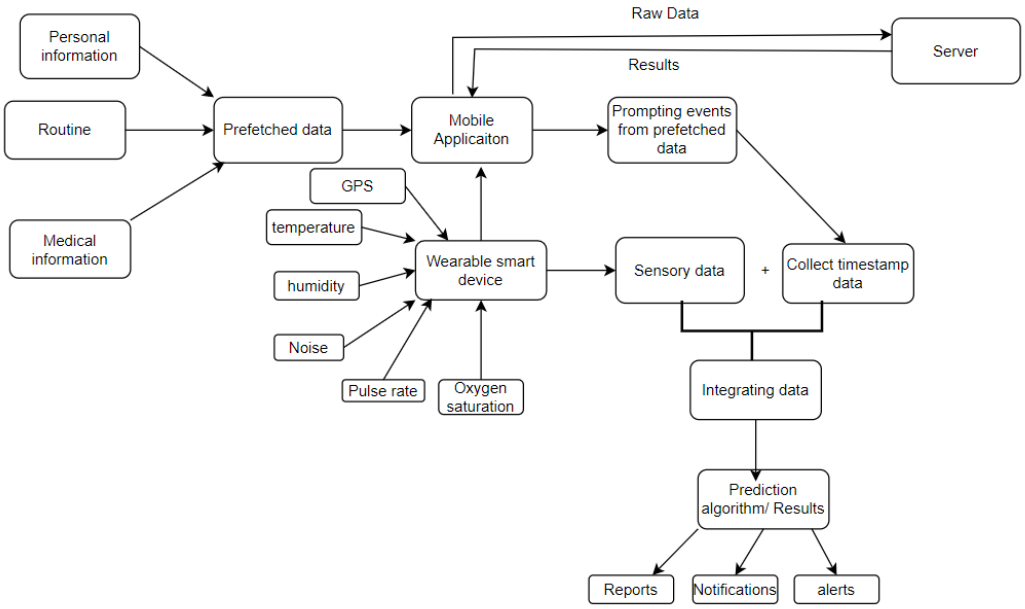
The device will be designed using a minimal number of sensors to reduce complexity and energy use ensuring all the critical data is collected in timely manner. Edge AI will be used to process data locally on the device allowing for immediate responses to changes in the child conditioning such as rise in temperature, emotional and behavioral disorders specially in children diagnosed with Autism, down syndrome.

Environmental sensors will monitor factors like temperature, humidity and noise, while physiological sensors will track pulse rate, oxygen saturation and GPS and accelerometers keep track of the child location and movement ensuring their safety.

Gathered information will be tracked on timely manner, providing comprehensive reports for the guardians about the growth, behavior of the child. The system can be further developed to automatic detection of child's emotional and physical changes with the environment. As of an example the device could automatically play calming music to calm them down while sending notifications to the parent. If the child leaves the safe area the device will alert the parent by providing notifications. Automatic notifications for notifying child's medicine routine can be done. Also, the device can be used as a log for keeping track of things that the child like to do, dislike to do, fear to do and using that data to make predictions in future. Recommendation of activities to engage in, creating daily routines, setting up breaks, assist in speech therapy, integrating therapy into daily life by making therapy fun with games, incorporating into routine can be added with the advancement of the device.

METHODOLOGY

High-Level Architecture



The wearable smart device will include sensors for measuring temperature, humidity, noise levels, pulse rate and oxygen saturation. These sensors will be embedded in the wearable device which will communicate wirelessly with the mobile application using Bluetooth or WIFI. The GPS will be included for location tracking while accelerometer will be used for detecting movement and fails. Sensor data will be collected in real time and processed locally on the device and periodically sent to the server for storage. Then analyzing the collected data a prediction algorithm can predict and suggest actions, send notifications to parents.

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